37 A radioactive isotope has a half-life of 8.0 days.

A detector close to a sample of this isotope gives a count rate of 200 counts per minute.

Without the source, the background count is 20 counts per minute.

What is the count rate **due to the source** after 8.0 days?

A 80 counts per minute

B 90 counts per minute

C 100 counts per minute

D 110 counts per minute